

EMBEDDED MACHINE LEARNING · PHD COURSE

SESSION 02

Data acquisition, preprocessing and feature engineering

What has already been lost before the model sees the data — and the cheapest representation that keeps what it needs.

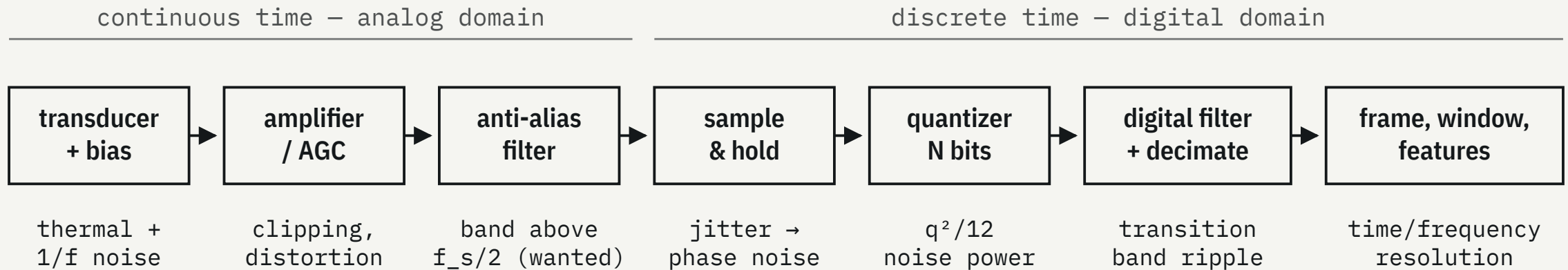
THE PRINCIPLE

The model can only ever recover what the chain preserved.

Two steps in this session are irreversible — aliasing and clipping. Everything else can be revisited later.

OVERVIEW

The chain, and what each stage destroys



every red annotation is information the model can never recover

The analog/digital boundary sits between the anti-alias filter and the sampler – and that placement is the whole point.

THE SENSORS YOU WILL MEET

Rate, resolution, interface, power

SENSOR	TYPICAL RATE	BITS	INTERFACE	ACTIVE POWER
MEMS microphone	16–48 kHz	16–24	PDM / I ² S	≈ 0.3–1 mW
Accelerometer	10 Hz – 6.4 kHz	12–16	I ² C / SPI	≈ 5–200 μW
Gyroscope	25 Hz – 6.4 kHz	16	I ² C / SPI	≈ 1–3 mW
PPG (optical HR)	25–500 Hz	16–20	I ² C	≈ 0.1–2 mW
ECG front end	125–1000 Hz	12–24	SPI / ADC	≈ 10–300 μW
Low-res camera	1–30 fps	8 / pixel	DCMI / MIPI	≈ 1–100 mW

The gyroscope costs 10–100× the accelerometer. And the sensor's configuration is part of the model's input spec.

01

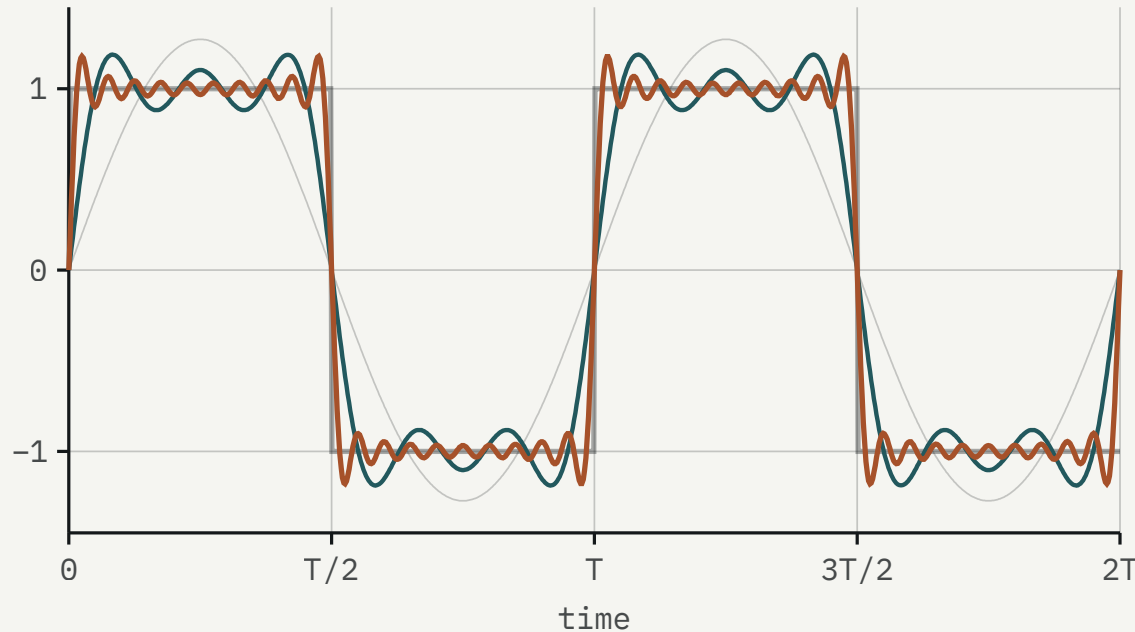
Signals in time and frequency

A primer — the language every later step is described in.

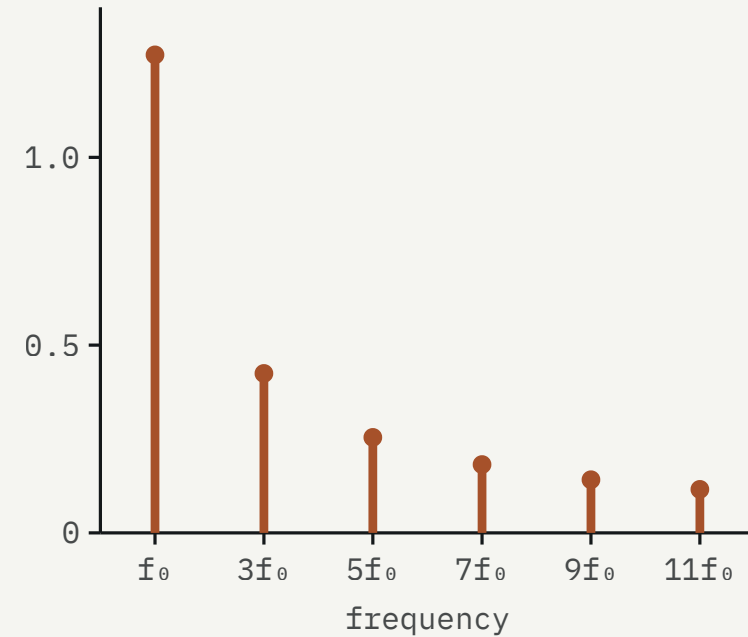
FOURIER

Any periodic signal is a sum of sinusoids

square wave = sum of odd sine harmonics



its spectrum: amplitude $4/(\pi k)$



PRIMER · THE DFT

Five lines on the discrete Fourier transform

$$X[k] = \sum_n x[n] \cdot e^{(-j2\pi kn/N)}, \quad k = 0 \dots N-1$$

$$\text{bin frequency} \quad f_k = k \cdot f_s / N \quad \text{resolution} \quad \Delta f = f_s / N$$

real signal: $N/2+1$ independent bins, highest frequency $f_s / 2$

$$\text{FFT cost} \approx N \cdot \log_2 N \quad (N = 512: \sim 4.6 \text{ k vs } N^2 \approx 262 \text{ k})$$

Each $X[k]$ measures how strongly the block correlates with a sinusoid at f_k . A longer block resolves finer frequency differences — at the price of time resolution.

PRIMER · CONVOLUTION

Why frequency is the natural language of the chain

- A linear filter = **convolution** with its impulse response: a sliding weighted sum.
- **Convolution in time = multiplication in frequency**: filtering scales each frequency.
- And the reverse: **sampling multiplies** by an impulse train → the spectrum is **copied** every f_s .
- Same word in Session 3: a CNN layer is a sliding weighted sum with **learned** weights.

01

Sampling and aliasing

Stated precisely, then demonstrated numerically.

THE THEOREM, AND WHAT IT REALLY SAYS

Sampling replicates the spectrum

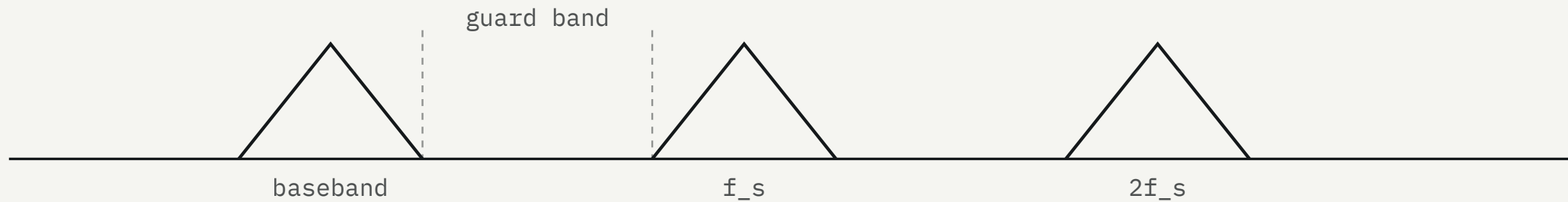
$$\text{Reconstructable} \iff f_s > 2B \quad (B = \text{highest frequency present})$$
$$f_{\text{alias}} = |f - \text{round}(f / f_s) \cdot f_s|$$

Sampling in time multiplies by an impulse train, which convolves the spectrum with an impulse train. The spectrum is periodically replicated. If $f_s < 2B$ the replicas overlap and ADD — and an addition whose terms were never observed apart cannot be undone.

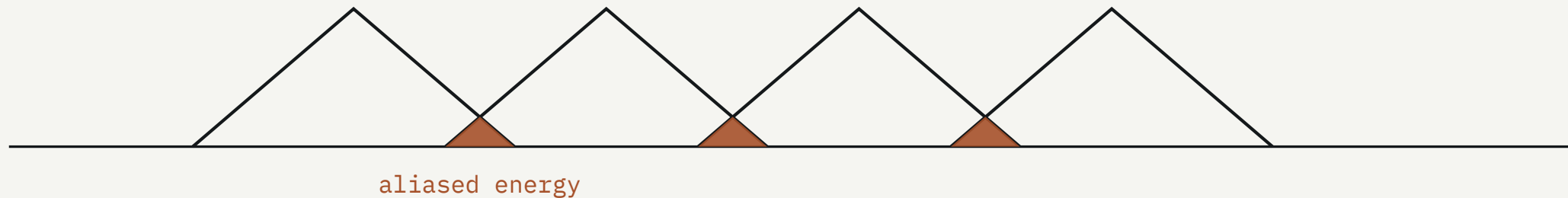
MECHANISM

Aliasing is spectral folding

$f_s > 2B$ – replicas separated, baseband recoverable



$f_s < 2B$ – replicas overlap; the sum in the shaded band cannot be undone



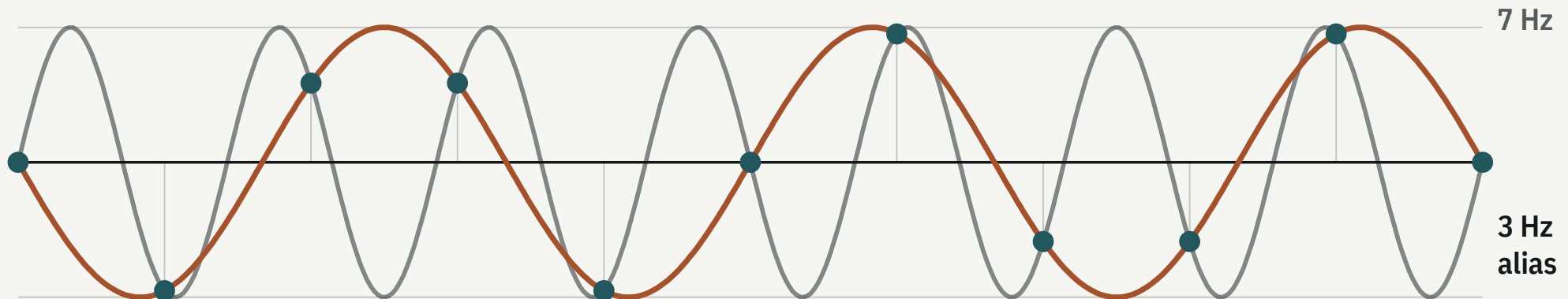
Replicas overlap → the observed spectrum is their **sum**.

COMPUTED, NOT ASSERTED

Same samples, two different signals

true 7 Hz · $f_s = 10$ Hz · Nyquist limit 5 Hz

teal dots = everything the converter ever sees



$$| 7 - \text{round}(7/10) \cdot 10 | = 3 \text{ Hz}$$

Same samples two different signals — the loss is already permanent

A 7 Hz tone sampled at 10 Hz produces **exactly** the sample set a 3 Hz tone would produce.

02

Quantization noise

Where the second irreversible loss happens — and how much of it you are allowed.

DERIVATION

From a uniform quantizer to $6.02 N + 1.76$ dB

$$q = V_{FS} / 2^N \quad \text{step size}$$

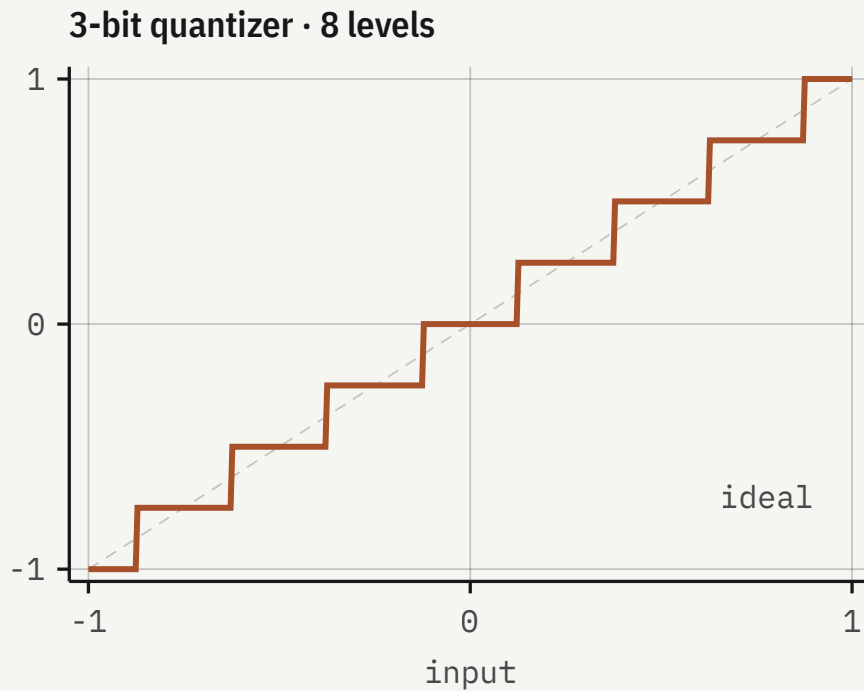
$$\sigma_e^2 = (1/q) \int_{-q/2}^{q/2} e^2 de = q^2/12$$

$$\text{SQNR} = 10 \log_{10} \left((V_{FS}^2/8) / (q^2/12) \right) \approx 6.02 N + 1.76 \quad [\text{dB}]$$

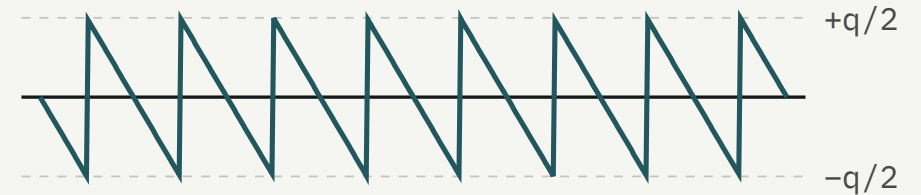
Assumption: the rounding error is uniform on $[-q/2, q/2]$ and independent of the signal. Good when the signal is busy relative to q — and false when it is not.

WHAT A QUANTIZER REALLY DOES

The error is a sawtooth, not noise



error $e = \hat{x} - x$



$$\sigma_e^2 = q^2/12$$

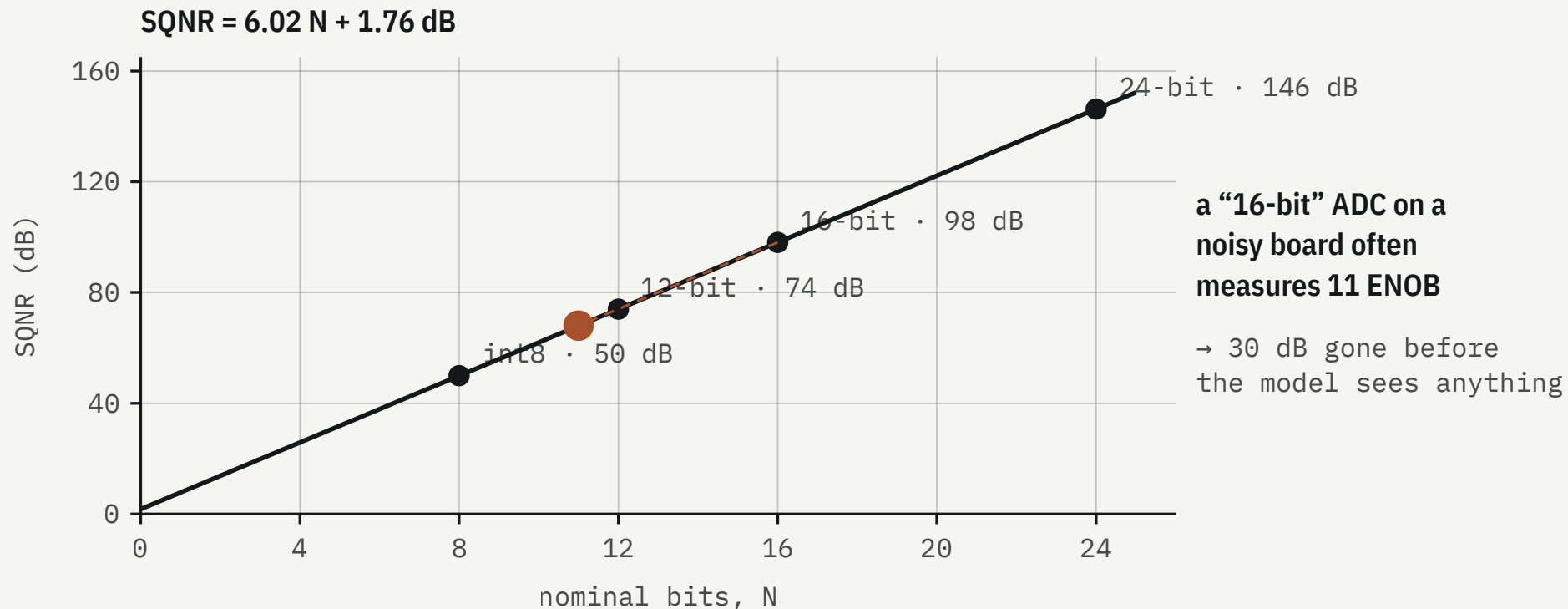
$$\rightarrow \text{SQNR} \approx 6.02 N + 1.76 \text{ dB}$$

Every flat tread is a set of inputs the converter cannot tell apart.

Every flat tread is a set of real inputs the converter cannot tell apart.

DESIGN AGAINST THIS, NOT THE LABEL

Bits are worth 6 dB – in both directions



$ENOB = (SINAD - 1.76) / 6.02$. A “16-bit” ADC on a noisy board often measures 11.

FOUNDATIONS

Three converter architectures, three currencies

ARCHITECTURE	MECHANISM	TYPICAL	IMPLICATION FOR YOU
SAR	binary search, one comparison per bit	8–18 bit kSPS–MSPS	The MCU default. Low latency, easy to multiplex across channels.
Sigma–delta	1-bit modulator at 64–256× oversampling + digital decimation	16–24 bit kSPS	Audio and precision sensing. Adds group delay — matters for event timestamps and sensor fusion.
Flash / pipeline	2 ^N –1 parallel comparators, or a pipeline of stages	6–14 bit MSPS–GSPS	Radar, ultrasound, RF. You will be bandwidth-limited downstream long before the converter is the problem.

Each buys resolution with a different currency: comparisons, time, or silicon area.

03

Noise and filtering

What actually limits your SNR — and it is rarely the quantizer.

CHOOSING A STRUCTURE

FIR versus IIR on an embedded target

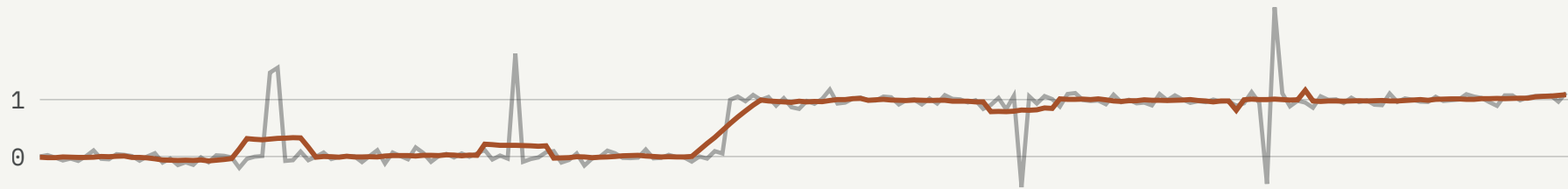
PROPERTY	FIR	IIR
Cost per output sample	M MACs (M taps)	$\approx 2 \times \text{order}$ (biquads)
State memory	M words	2 words per biquad
Phase	exactly linear if symmetric	non-linear; group delay varies
Stability	always stable	poles must stay inside the unit circle
Fixed-point failure	accumulator overflow	coefficient quantization moves poles; limit cycles
Use for	decimation, matched filtering, anything phase-sensitive	cheap band-limiting, DC blocking, envelopes

For a given magnitude spec an IIR is typically **an order of magnitude cheaper**. You pay in phase linearity and numerical fragility.

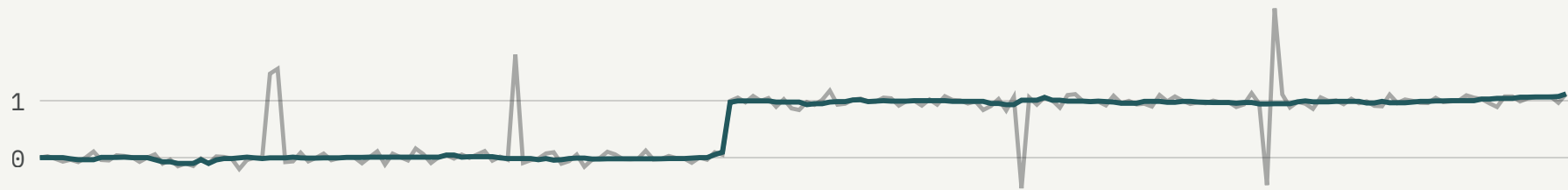
TWO CHEAP FILTERS

Moving average versus median

moving average, 9 taps — linear, cheap, smears spikes and edges



median, 9 taps — non-linear, removes spikes, keeps the edge



grey: raw samples (step + Gaussian noise + six impulsive spikes)

SAMPLING RATE

Halving f_s halves almost everything downstream

- ADC energy per conversion $\times$ **rate**
- Filter MACs per sample $\times$ **rate**
- Frames per second $\times$ feature cost
- Inferences per second $\times$ model cost
- **But:** only free of accuracy loss if the discarded band carries no class-discriminative information — **test it, do not assume it.**

Keyword spotting: 16 kHz conventional, 8 kHz often enough. Bearing faults: the diagnostic energy may sit above 10 kHz and the same reasoning gives the opposite answer.

BREAK

Five minutes.

Next: datasets and preprocessing, then features.

02

Datasets, labels and preprocessing

The most expensive and least reusable part of most projects.

BUILDING A DATASET

Coverage, labels, rarity

- **Coverage beats volume:** users, mountings, devices, environments.
- **Label granularity** is a design decision: per recording, per window, exact boundaries.
- **Rare events** need targeted collection and a diverse background class.
- **Preprocessing** — sync, resample, gaps, calibration — must exist **identically** in firmware.
- **Augment:** noise, gain, shifts, rotations, SpecAugment. Free at inference.

NORMALISATION

Scale features before most models see them

raw values

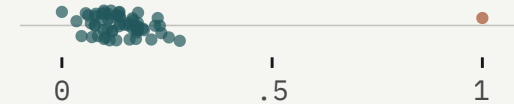
RMS level, mV



zero-crossing rate



spectral peak, Hz

z-score: $(x - \mu)/\sigma$ min-max: $(x - \min)/(\max - \min)$ 

Raw: features 1 and 2 vanish next to feature 3 – a distance-based model (k-NN, RBF-SVM, k-means) effectively ignores them.

Min-max: one outlier (copper) squeezes the other 59 values of feature 3 into a fifth of the range. Fit all scaling statistics on the training set only.

THE COMMONEST INVALID RESULT

Split by subject, device or recording — never by window.

Overlapping windows from one recording are near-copies. A random split measures memorisation.

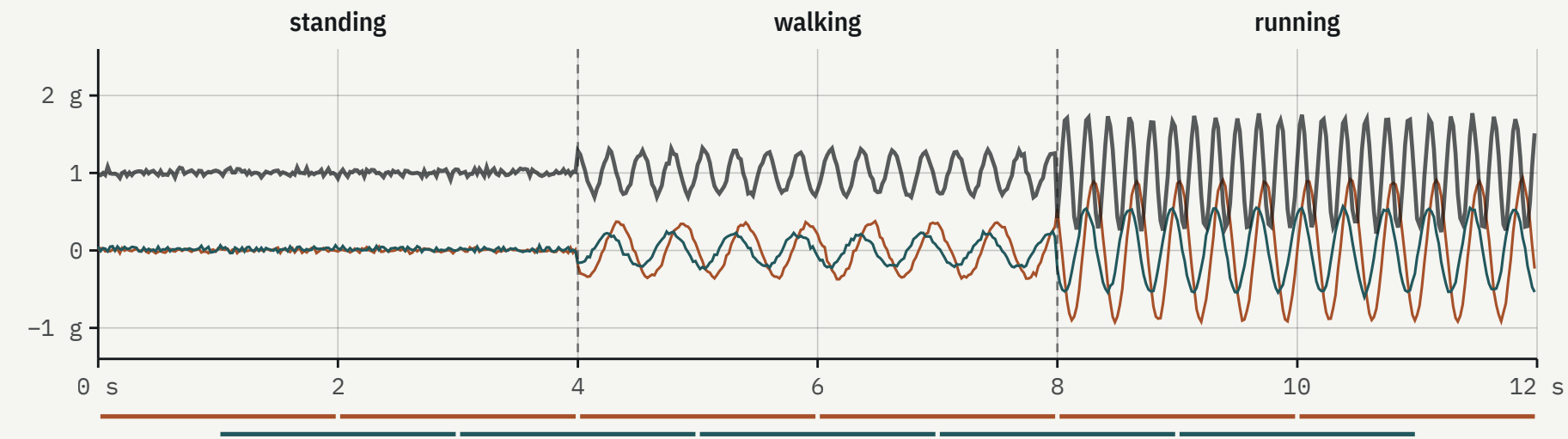
BRIDGE TO SESSION 2

Framing: the model consumes frames, not streams

- Frame length **L**, hop **H** $\rightarrow$ overlap $1 - H/L$, frame rate f_s/H
- Frame **rate** sets the inference rate, and therefore the energy
- Frame **length** sets the lowest resolvable frequency, f_s/L
- Hop sets the temporal precision of an event onset
- **Overlap is not free** — 50 % overlap doubles the number of inferences

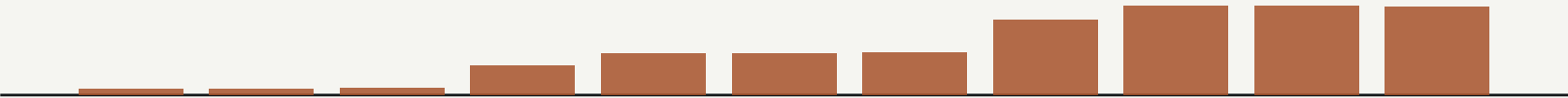
SEGMENTATION

From a stream to examples



windows: 2 s (100 samples) with 50 % overlap → one feature vector every second

$\sigma(|a|)$



one simple feature per window – the standard deviation of $|a|$ – already separates the three activities

METHOD

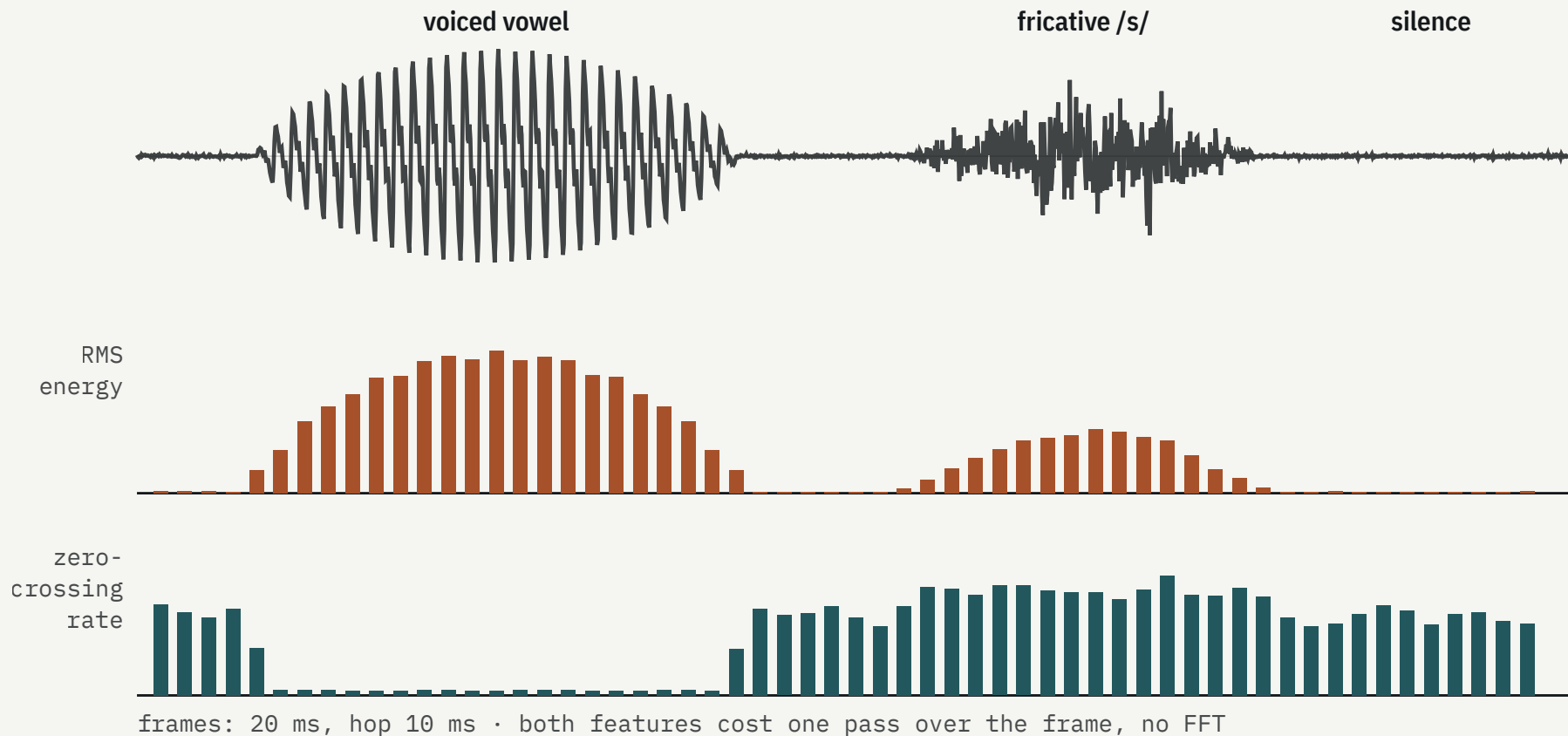
Three questions to ask of any representation

- **What does it discard** — and is the discarded part class-discriminative?
- **What invariances does it build in?** A magnitude spectrogram discards phase, buying shift invariance. If your task needs inter-channel phase, you just deleted the signal.
- **What does it cost**, compared with the model it feeds?

If the front end costs 40 % of the budget, a learned front end on raw samples may be cheaper overall.

TIME-DOMAIN FEATURES

Energy and zero-crossing rate: a voice-activity detector for free



01

The short-time Fourier transform

Windows, leakage, and a bound you cannot tune your way around.

DEFINITION

Windowed transforms of overlapping segments

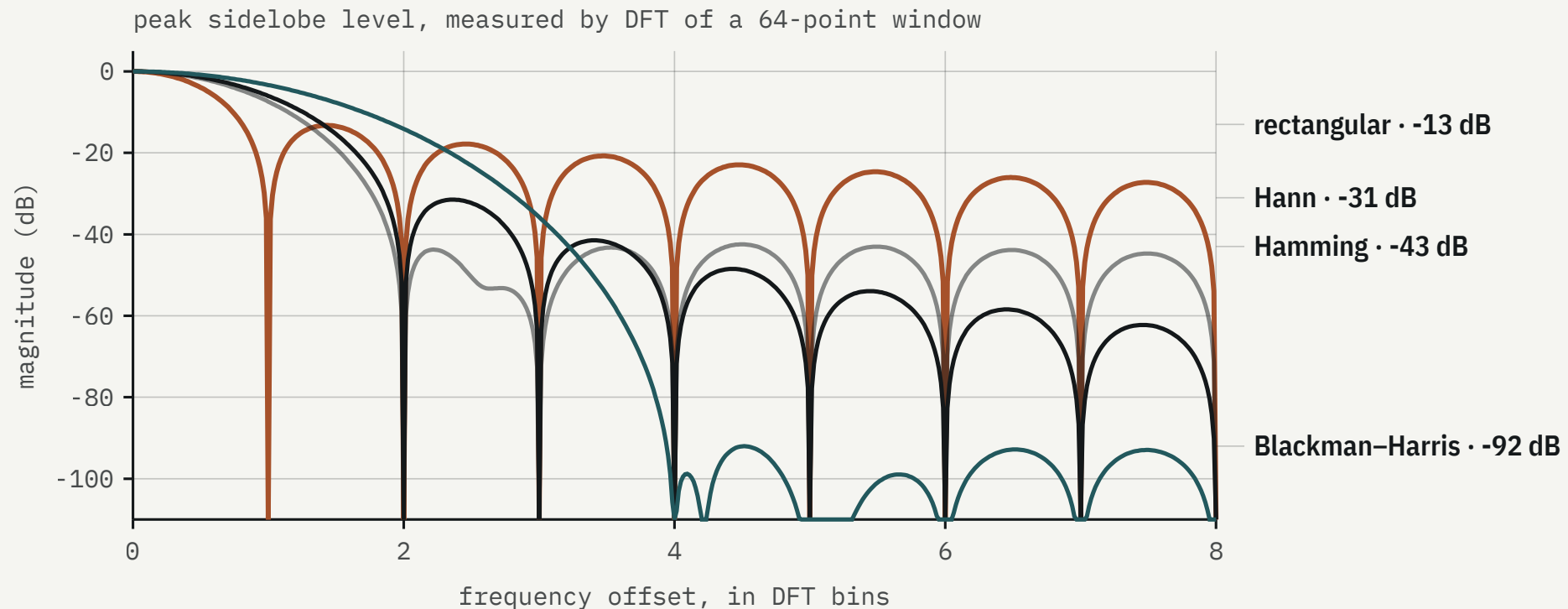
$$X[m, k] = \sum_n x[n] \cdot w[n - mH] \cdot \exp(-j2\pi kn/L)$$

$$\text{spectrogram} = |X[m, k]|^2$$

The Fourier transform assumes stationarity over its whole support. Speech, vibration and gesture signals are not stationary — so we apply it to short segments in which stationarity is approximately true.

MEASURED, NOT ASSERTED

Narrow main lobe or low sidelobes – not both



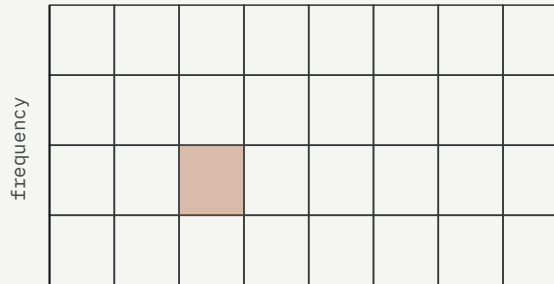
Rectangular -13 dB · Hann -31 · Hamming -43 · Blackman-Harris -92.

THE BOUND

Every tile has the same area

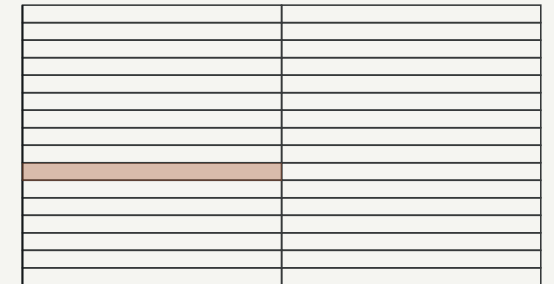
- $\Delta f \approx f_s / L$ $\Delta t = L / f_s$ $\Delta t \cdot \Delta f \approx 1$
- You cannot resolve a 20 ms transient **and** a 10 Hz frequency difference in the same analysis.
- Overlap increases the **frame rate**, not the resolution.
- The smearing is set by L . Students conflate these constantly.

short window $L = 128$ · Δt small, Δf large



time

long window $L = 1024$ · Δt large, Δf small



time

every tile has the same area – the window chooses its shape, not its size

02

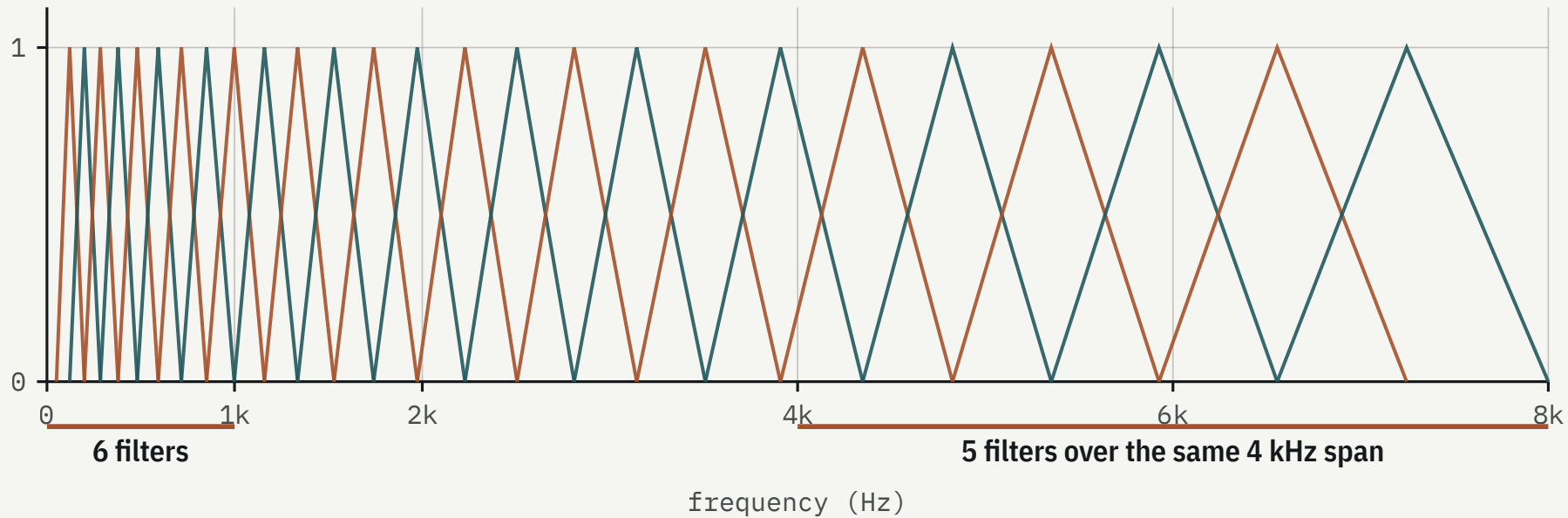
From spectrogram to mel and MFCC

Four steps, each of which had a reason — and one of which no longer applies.

PERCEPTUAL WARPING

Resolution spent where hearing has it

26 triangular filters, equally spaced on the mel axis, drawn on a linear Hz axis



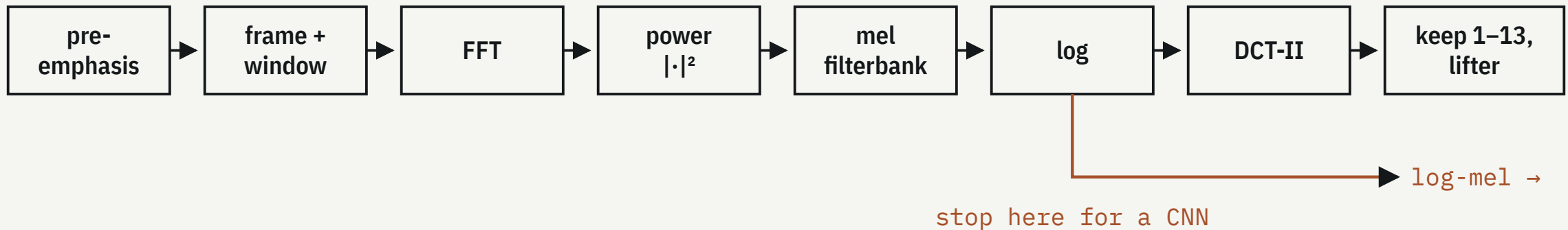
$$m(f) = 2595 \cdot \log_{10}(1 + f / 700)$$

$m(f) = 2595 \cdot \log_{10}(1 + f/700)$. Six filters below 1 kHz; five across the whole 4–8 kHz span.

THE PIPELINE, AND THE BRANCH THAT MATTERS

Where to stop for a CNN

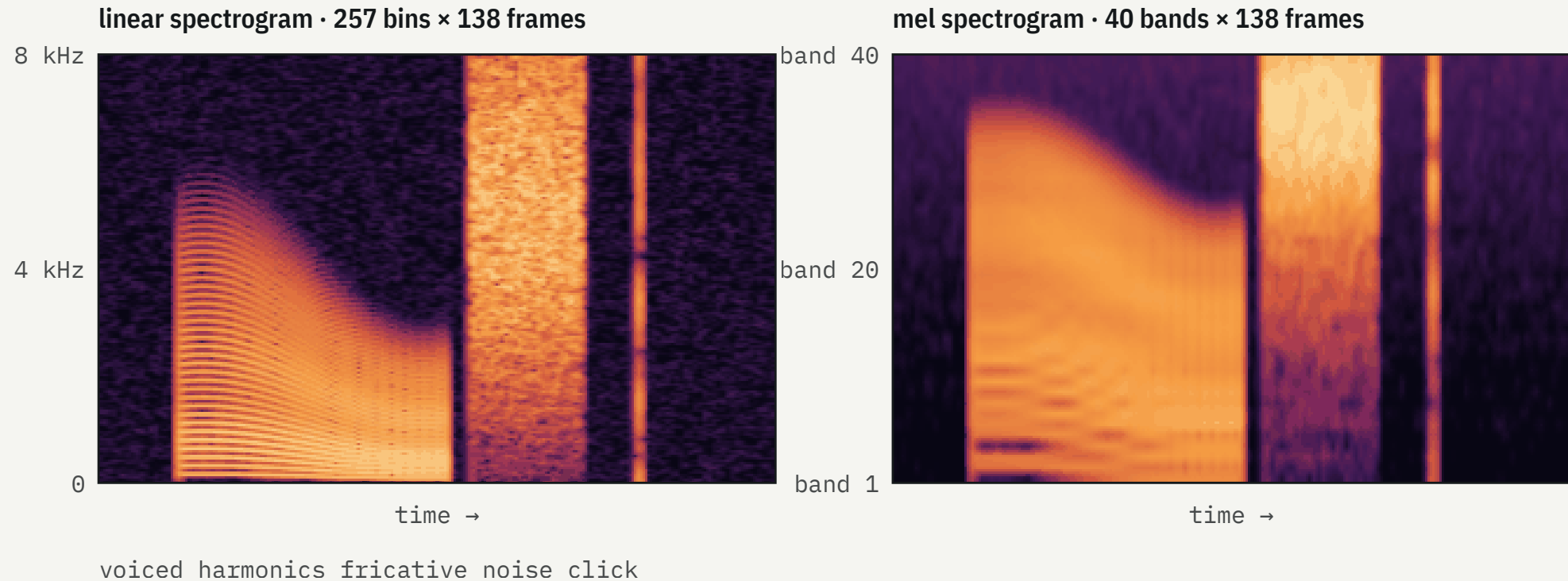
cost per frame $\approx O(L \log L)$ for the FFT + $B \cdot L/2$ for the filterbank + $B \log B$ for the DCT
→ the filterbank is sparse in practice: each bin touches at most 2 triangles



The DCT existed to decorrelate for **diagonal-covariance GMMs**. A CNN has no such requirement – and the DCT destroys the local structure convolutions exploit.

THE SAME SOUND, TWO REPRESENTATIONS

6.4× fewer numbers, nearly all the structure



Voiced harmonics, a fricative burst, a click – all still legible after mel compression.

PRICE THE FRONT END

Front end against model, order of magnitude

FRONT END	COST / FRAME	OUTPUT DIM	NOTE
Time-domain statistics, 3-axis IMU, 128 samples	$\sim 10^3$ ops	~ 30	Runs in the sensor ISR
512-point real FFT	$\sim 10^4$ MACs	257	CMSIS-DSP radix-2/4
40-band log-mel from that FFT	$\sim 1.5 \times 10^4$	40	Filterbank is sparse
13 MFCC + Δ + $\Delta\Delta$	$\sim 1.6 \times 10^4$	39	Adds a small DCT
Small DS-CNN over a 49×10 map	$\sim 2.7 \times 10^6$	—	Per inference $\approx 4\times$ the front end of a whole second (49 frames $\approx 7 \times 10^5$)

The comparison flips for very small models and high sample rates. **Measure, do not assume.**

03

Selection and reduction

Which features survive — and the hygiene rule that decides whether a result is valid.

THREE FAMILIES

Filter, wrapper, embedded

Filter — score each feature alone

- Pearson ρ : linear dependence only
- Mutual information: any dependence, but must be estimated
- Cheap; blind to redundancy
- Two perfectly correlated informative features both score high, both get kept

Wrapper & embedded

- RFE: fit, rank, drop the weakest, repeat
- Expensive, but accounts for interaction and redundancy
- Lasso: L1 corner geometry drives coefficients to exactly zero
- Tree importances — prefer permutation over impurity, which is biased toward high-cardinality features

THE HYGIENE RULE

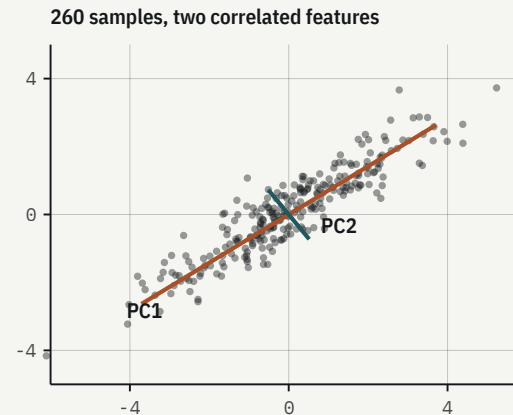
Feature selection is part of the model, so it happens *inside* each cross-validation fold.

Selecting on the full dataset and then cross-validating the classifier leaks label information — routinely several accuracy points, and far more when features outnumber samples. The same applies to normalisation statistics, PCA bases and class balancing.

DIMENSIONALITY REDUCTION

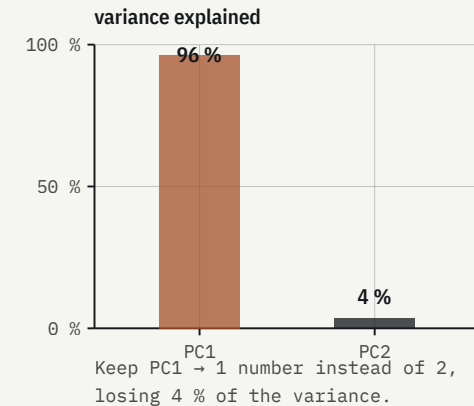
PCA, and the cost nobody counts

- Maximise $w^T \Sigma w$ subject to $\|w\|=1 \rightarrow \Sigma w = \lambda w$: the eigenvectors of the covariance matrix.
- Compute it by **SVD of the centred data**, not by forming Σ — which squares the condition number.
- On device it is one stored $D \times d$ matrix multiply.
- **Count its bytes.** 512 $\rightarrow$ 32 is 16 384 stored coefficients — often more than the classifier you were shrinking.



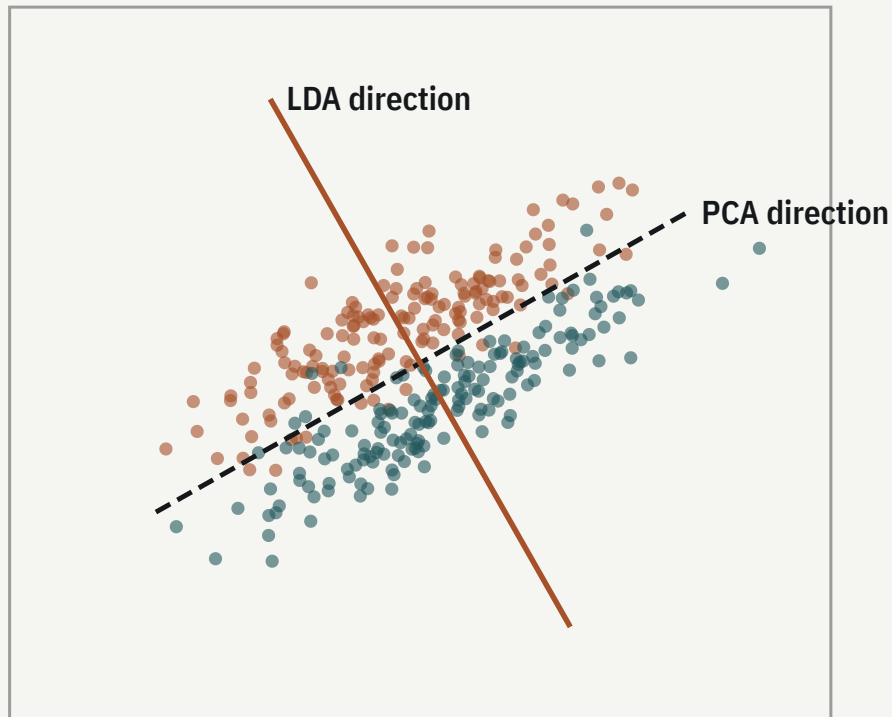
PCA finds the eigenvectors of the covariance matrix; the eigenvalue is the variance along each.

On device it is one stored matrix multiply — count its bytes before assuming it saves any.

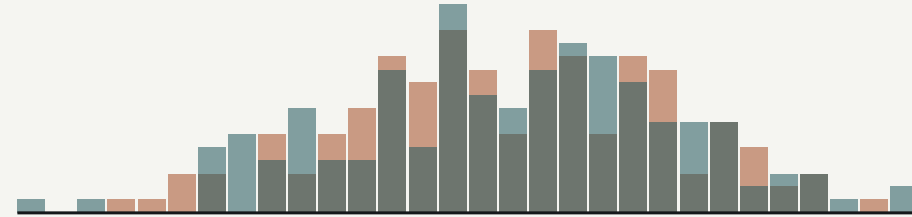


SUPERVISED PROJECTION

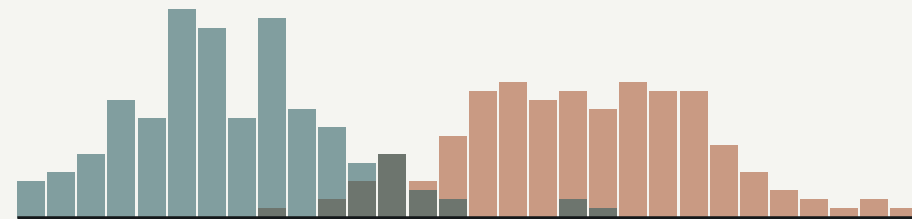
Greatest variance is not greatest separation



projection on the PCA direction — classes overlap



projection on the LDA direction — classes separate

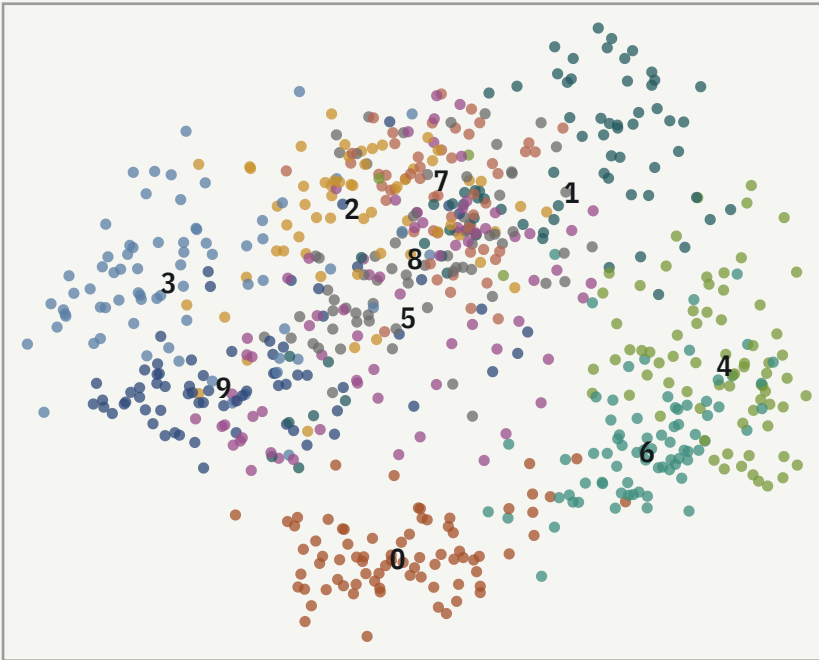


PCA ignores labels; LDA uses them (supervised).

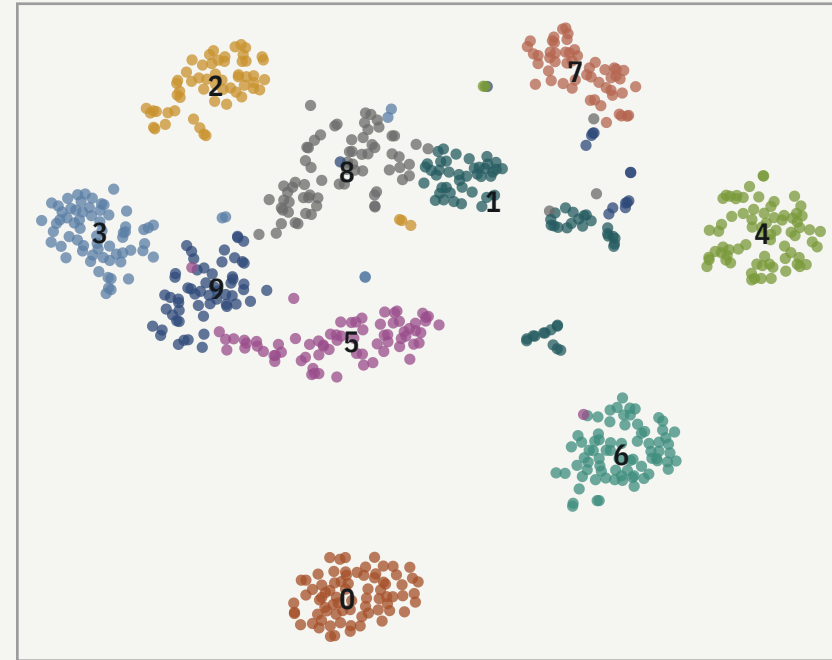
LOOKING AT DATA

t-SNE is for looking, not for deploying

PCA — linear, global, 2 of 64 dimensions



t-SNE — non-linear, local neighbourhoods



scikit-learn digits dataset (8×8 images = 64 features), 700 random samples; colour and numeral = true class.
t-SNE axes mean nothing and between-cluster distances are not preserved: use it to look.

BEFORE SESSION 3

Reading

Y. Zhang, N. Suda, L. Lai, V. Chandra, “Hello Edge: Keyword Spotting on Microcontrollers,” 2017. arXiv:1711.07128.

Why: A complete embedded pipeline — MFCC front end, several model families, memory and operation budgets — in twelve pages. Session 3 uses its numbers.

No ML background? Work through Part A of the Session 3 web chapter before the lecture.

IF YOU REMEMBER ONE THING

Decide what the model must distinguish, then work backwards: bandwidth sets the sampling rate, dynamic range sets the bits, event duration sets the window, and the cheapest representation that keeps the distinguishing information is the right one.

BEFORE YOU GO

Think about these

- Q1 A colleague samples an accelerometer at 25 Hz “because human motion is below 10 Hz”. What must you know about the sensor's internal filter before agreeing?
- Q2 Name a task for which discarding phase destroys the information the model needs.
- Q3 Your accuracy falls from 97 % to 81 % with a subject-wise split. What do you report?